

Overview

This document explains the comprehensive exception handling strategy implemented in the CivilControl Backend system, providing consistent error responses, proper HTTP status codes, and integration with i18n for localized error messages.

Architecture

Core Components

1. **GlobalExceptionHandler** - Central exception handling
2. **Custom Exception Hierarchy** - Specific business exceptions
3. **ErrorResponse** - Standardized error response format
4. **i18n Integration** - Translated error messages
5. **Validation Error Handling** - Bean Validation integration

Exception Hierarchy

Base Exception Classes

RuntimeException └─ NotFoundException (404) └─ UserNotFoundException └─ VehicleNotFoundException └─ BuildingNotFoundException └─ ... (other NotFoundException exceptions) └─ AlreadyExistsException (409) └─ UserAlreadyExistsException └─ VehicleAlreadyExistsException └─ BuildingAlreadyExistsException └─ ... (other AlreadyExists exceptions) └─ NotValidException (400) └─ UserNotValidException └─ VehicleNotValidException └─ BuildingNotValidException └─ ... (other NotValid exceptions) └─ DataConflictException (409) └─ VehicleDataConflictException └─ BuildingDataConflictException └─ ... (other DataConflict exceptions)

Custom Exceptions

1. NotFoundException (404)

Purpose: Resource not found in database

```
package CivilControl.backEnd.exception; public abstract class NotFoundException extends RuntimeException { public NotFoundException(String message) { super(message); } }
```

Example:

```
public class VehicleNotFoundException extends NotFoundException { public
VehicleNotFoundException(Long id) {
super(MessageSourceHelper.getMessageStatic("vehicle.notFound", id)); } public
VehicleNotFoundException(String licensePlate) {
super(MessageSourceHelper.getMessageStatic( "vehicle.notFoundByLicensePlate",
licensePlate)); } }
```

Usage:

```
Vehicle vehicle = vehicleRepository.findById(id) .orElseThrow(() -> new
VehicleNotFoundException(id));
```

2. AlreadyExistsException (409)

Purpose: Resource already exists (duplicate)

```
public class VehicleAlreadyExistsException extends RuntimeException { public
VehicleAlreadyExistsException(String message) { super(message); } }
```

Usage:

```
if (vehicleRepository.existsByLicensePlate(licensePlate)) { throw new
VehicleAlreadyExistsException( messageSourceHelper.getMessage(
"vehicle.licensePlate.alreadyExists", licensePlate)); }
```

3. NotValidException (400)

Purpose: Business logic validation failure

```
public class VehicleNotValidException extends RuntimeException { public
VehicleNotValidException(String message) { super(message); } }
```

Usage:

```
if (dto.year() < 2000) { throw new VehicleNotValidException(
messageSourceHelper.getMessage("vehicle.year.tooOld")); }
```

4. DataConflictException (409)

Purpose: Data integrity conflict

```
public class VehicleDataConflictException extends RuntimeException { public
VehicleDataConflictException(String message) { super(message); } public
VehicleDataConflictException(String message, Throwable cause) { super(message, cause); } }
```

Usage:

```
try { vehicleRepository.save(vehicle); } catch (DataIntegrityViolationException e) { throw new
VehicleDataConflictException( messageSourceHelper.getMessage("vehicle.dataConflict"), e); }
```

Global Exception Handler

GlobalExceptionHandler.java

Located in: src/main/java/CivilControl/backEnd/config/GlobalExceptionHandler.java

Purpose: Centralized exception handling for all controllers

```
@RestControllerAdvice @Slf4j public class GlobalExceptionHandler { @Autowired private
MessageSourceHelper messageSourceHelper; /** * Handle NotFoundException (404) */
@ExceptionHandler(NotFoundException.class) @ResponseStatus(HttpStatus.NOT_FOUND)
public ErrorResponse handleNotFoundException(NotFoundException ex) { log.warn("Not found:
{}", ex.getMessage()); return ErrorResponse.builder() .status(HttpStatus.NOT_FOUND.value())
.message(ex.getMessage()) .timestamp(LocalDateTime.now()) .build(); } /** * Handle
AlreadyExistsException (409) */ @ExceptionHandler({AlreadyExistsException.class,
DataConflictException.class}) @ResponseStatus(HttpStatus.CONFLICT) public ErrorResponse
handleConflictException(RuntimeException ex) { log.warn("Conflict: {}", ex.getMessage());
return ErrorResponse.builder() .status(HttpStatus.CONFLICT.value())
.message(ex.getMessage()) .timestamp(LocalDateTime.now()) .build(); } /** * Handle
NotValidException (400) */ @ExceptionHandler(NotValidException.class)
@ResponseStatus(HttpStatus.BAD_REQUEST) public ErrorResponse
handleNotValidException(NotValidException ex) { log.warn("Validation error: {}",
ex.getMessage()); return ErrorResponse.builder() .status(HttpStatus.BAD_REQUEST.value())
.message(ex.getMessage()) .timestamp(LocalDateTime.now()) .build(); } /** * Handle Bean
Validation errors (400) */ @ExceptionHandler(MethodArgumentNotValidException.class)
@ResponseStatus(HttpStatus.BAD_REQUEST) public ErrorResponse
handleValidationException(MethodArgumentNotValidException ex) { log.warn("Validation errors
in request"); Map<String, String> errors = new HashMap<>();
ex.getBindingResult().getFieldErrors().forEach(error -> { errors.put(error.getField(),
error.getDefaultMessage()); }); return ErrorResponse.builder()
.status(HttpStatus.BAD_REQUEST.value())
.message(messageSourceHelper.getMessage("validation.error"))
.timestamp(LocalDateTime.now()) .errors(errors) .build(); } /** * Handle authentication errors
(401) */ @ExceptionHandler({InvalidCredentialsException.class, InvalidTokenException.class})
@ResponseStatus(HttpStatus.UNAUTHORIZED) public ErrorResponse
handleAuthenticationException(RuntimeException ex) { log.warn("Authentication error: {}",
ex.getMessage()); return ErrorResponse.builder() .status(HttpStatus.UNAUTHORIZED.value())
.message(ex.getMessage()) .timestamp(LocalDateTime.now()) .build(); } /** * Handle
authorization errors (403) */ @ExceptionHandler(AccessDeniedException.class)
@ResponseStatus(HttpStatus.FORBIDDEN) public ErrorResponse
handleAccessDeniedException(AccessDeniedException ex) { log.warn("Access denied: {}",
ex.getMessage()); return ErrorResponse.builder() .status(HttpStatus.FORBIDDEN.value())
.message(messageSourceHelper.getMessage("access.denied"))
.timestamp(LocalDateTime.now()) .build(); } /** * Handle unexpected errors (500) */
@ExceptionHandler(Exception.class)
```

```
@ResponseStatus(HttpStatus.INTERNAL_SERVER_ERROR) public ErrorResponse
handleGenericException(Exception ex) { log.error("Unexpected error", ex); return
ErrorResponse.builder().status(HttpStatus.INTERNAL_SERVER_ERROR.value())
.message(messageSourceHelper.getMessage("error.unexpected", ex.getMessage()))
.timestamp(LocalDateTime.now()).build(); }
```

Error Response Format

ErrorResponse.java

```
@Data @Builder @NoArgsConstructor @AllArgsConstructor public class ErrorResponse {
private int status; private String message; private LocalDateTime timestamp;
@JsonInclude(JsonInclude.Include.NON_NULL) private Map<String, String> errors; // For
validation errors }
```

Response Examples

1. Not Found (404)

```
{ "status": 404, "message": "No se encontro el vehiculo con ID 123", "timestamp":
"2026-01-20T15:30:00" }
```

2. Validation Error (400)

```
{ "status": 400, "message": "Error de validacion", "timestamp": "2026-01-20T15:30:00", "errors":
{ "licensePlate": "La patente es obligatoria", "year": "El año debe ser mayor a 2000" } }
```

3. Conflict (409)

```
{ "status": 409, "message": "Ya existe un vehiculo con la patente: ABC123", "timestamp":
"2026-01-20T15:30:00" }
```

4. Unauthorized (401)

```
{ "status": 401, "message": "Credenciales invalidas", "timestamp": "2026-01-20T15:30:00" }
```

5. Forbidden (403)

```
{ "status": 403, "message": "No tiene permisos para acceder a este recurso", "timestamp":
```

"2026-01-20T15:30:00" }

6. Internal Server Error (500)

```
{ "status": 500, "message": "Ha ocurrido un error inesperado: Connection timeout", "timestamp":  
"2026-01-20T15:30:00" }
```

HTTP Status Code Mapping

Status Code	
Exception Type	
Usage	
400	
NotValidException	
Business validation failure	
400	
MethodArgumentNotValidException	
Bean Validation failure	
401	
InvalidCredentialsException	
Wrong username/password	
401	
InvalidTokenException	
Invalid/expired JWT token	
403	
AccessDeniedException	
Insufficient permissions	
404	
NotFoundException	
Resource not found	
409	
AlreadyExistsException	
Duplicate resource	
409	
DataConflictException	
Data integrity violation	
500	
Exception	
Unexpected error	

Bean Validation Integration

Validation Annotations

```
public record VehicleDTO( @NotBlank(message = "{vehicle.licensePlate.required}")
    @Pattern(regexp = "^([A-Z]{3}\\d{3})$", message = "{vehicle.licensePlate.pattern}") String
    licensePlate, @NotNull(message = "{vehicle.brand.required}") @Size(min = 2, max = 50,
    message = "{vehicle.brand.size}") String brand, @NotNull(message = "{vehicle.year.required}")
    @Min(value = 2000, message = "{vehicle.year.min}") @Max(value = 2100, message =
    "{vehicle.year.max}") Integer year, @NotNull(message = "{vehicle.vehicleType.required}")
    VehicleType vehicleType, @Email(message = "{contact.email.invalid}") String contactEmail ) {}
```

Custom Validators

```
@Target({ElementType.FIELD}) @Retention(RetentionPolicy.RUNTIME)
@Constraint(validatedBy = LicensePlateValidator.class) public @interface ValidLicensePlate {
    String message() default "{vehicle.licensePlate.invalid}"; Class<?>[] groups() default {}; Class<?
    extends Payload>[] payload() default {}; } public class LicensePlateValidator implements
    ConstraintValidator<ValidLicensePlate, String> { @Override public boolean isValid(String value,
    ConstraintValidatorContext context) { if (value == null) return true; // Custom validation logic
    return value.matches("^([A-Z]{3}\\d{3})$") || value.matches("^([A-Z]{2}\\d{3}[A-Z]{2})$"); } }
```

Usage:

```
public record VehicleDTO( @ValidLicensePlate String licensePlate, // ... other fields ) {}
```

Service Layer Exception Handling

Exception Throwing Pattern

```
@Service @RequiredArgsConstructor public class VehicleService { private final
    VehicleRepository vehicleRepository; private final MessageSourceHelper
    messageSourceHelper; public VehicleResponseDTO getVehicleById(Long id) { // Throw
    NotFoundException if not found Vehicle vehicle =
    vehicleRepository.findByIdAndDeletedFalse(id).orElseThrow(() -> new
    VehicleNotFoundException(id)); return vehicleMapper.toResponseDto(vehicle); } public
    VehicleResponseDTO createVehicle(VehicleDTO dto) { // Check for duplicates if
    (vehicleRepository.existsByLicensePlate(dto.licensePlate())) { throw new
    VehicleAlreadyExistsException( messageSourceHelper.getMessage(
    "vehicle.licensePlate.alreadyExists", dto.licensePlate())); } // Business validation
    validateVehicle(dto); // Handle data integrity violations try { Vehicle vehicle =
    vehicleMapper.toEntity(dto); Vehicle saved = vehicleRepository.save(vehicle); return
    vehicleMapper.toResponseDto(saved); } catch (DataIntegrityViolationException e) { throw new
    VehicleDataConflictException( messageSourceHelper.getMessage("vehicle.dataConflict"), e); } }
    private void validateVehicle(VehicleDTO dto) { if (dto.year() < 2000) { throw new
```

```
VehicleNotValidException( messageSourceHelper.getMessage("vehicle.year.tooOld")); } if  
(dto.vehicleType() == VehicleType.CAR && dto.truckEquipment() != null) { throw new  
VehicleNotValidException( messageSourceHelper.getMessage("vehicle.car.noEquipment")); } }
```

Best Practices

1. Use Specific Exception Classes

```
// Good - Specific exception throw new VehicleNotFoundException(id); // Avoid - Generic  
exception throw new RuntimeException("Vehicle not found: " + id);
```

2. Use i18n for Messages

```
// Good - i18n message throw new VehicleNotFoundException(  
messageSourceHelper.getMessage("vehicle.notFound", id)); // Avoid - Hardcoded message  
throw new VehicleNotFoundException("Vehicle not found with ID: " + id);
```

3. Log Appropriately

```
// Info/Warn for expected exceptions log.warn("Vehicle not found: {}", id); // Error for unexpected  
exceptions log.error("Unexpected error processing vehicle", exception);
```

4. Don't Swallow Exceptions

```
// Good - Propagate or handle properly try { processVehicle(vehicle); } catch (Exception e) {  
log.error("Error processing vehicle", e); throw new VehicleProcessingException("Failed to  
process vehicle", e); } // Avoid - Swallowing exception try { processVehicle(vehicle); } catch  
(Exception e) { // Silent failure - BAD! }
```

5. Use @ExceptionHandler in Controllers (Optional)

For controller-specific exceptions:

```
@RestController @RequestMapping("/api/v1/vehicles") public class VehicleController {  
    @ExceptionHandler(InvalidVehicleTypeException.class)  
    @ResponseStatus(HttpStatus.BAD_REQUEST) public ErrorResponse  
    handleInvalidVehicleType(InvalidVehicleTypeException ex) { return ErrorResponse.builder()  
    .status(400) .message(ex.getMessage()) .timestamp(LocalDate.now()) .build(); } }
```

Testing Exception Handling

Unit Tests

```
@SpringBootTest class VehicleServiceTest { @Autowired private VehicleService vehicleService;
@Test void shouldThrowNotFoundException_whenVehicleDoesNotExist() { // Given Long
nonExistentId = 999L; // When & Then assertThrows(VehicleNotFoundException.class, () ->
vehicleService.getVehicleById(nonExistentId)); } @Test void
shouldThrowAlreadyExistsException_whenDuplicateLicensePlate() { // Given VehicleDTO dto =
createVehicleDTO("ABC123"); vehicleService.createVehicle(dto); // Create first vehicle // When
& Then assertThrows(VehicleAlreadyExistsException.class, () ->
vehicleService.createVehicle(dto)); // Try to create duplicate } }
```

Integration Tests

```
@SpringBootTest(webEnvironment = WebEnvironment.RANDOM_PORT) class
VehicleControllerIntegrationTest { @Autowired private TestRestTemplate restTemplate; @Test
void shouldReturn404_whenVehicleNotFound() { // When ResponseEntity<ErrorResponse>
response = restTemplate .getForEntity("/api/v1/vehicles/999", ErrorResponse.class); // Then
assertEquals(HttpStatus.NOT_FOUND, response.getStatusCode());
assertNotNull(response.getBody()); assertEquals(404, response.getBody().getStatus()); } @Test
void shouldReturn400_whenValidationFails() { // Given VehicleDTO invalidDto = new
VehicleDTO( "", // Empty license plate null, // Null brand null, // Null model 1990, // Year too old
null, null, 1L, null, null ); // When ResponseEntity<ErrorResponse> response = restTemplate
.postForEntity("/api/v1/vehicles", invalidDto, ErrorResponse.class); // Then
assertEquals(HttpStatus.BAD_REQUEST, response.getStatusCode());
assertNotNull(response.getBody()); assertNotNull(response.getBody().getErrors()); } }
```

Troubleshooting

Issue: Exception not caught by GlobalExceptionHandler

Solutions:

1. Ensure exception extends appropriate base class
2. Check if `@RestControllerAdvice` is in component scan path
3. Verify exception is not caught elsewhere

Issue: Wrong HTTP status code

Solutions:

1. Check `@ResponseStatus` annotation
2. Verify exception handler mapping
3. Ensure correct exception type is thrown

Issue: Validation messages not translated

Solutions:

1. Check message key exists in messages_es.properties
2. Verify i18n configuration
3. Ensure {} brackets around message key

Issue: Stack trace exposed to client

Solutions:

1. Don't include exception details in error response
2. Log full exception server-side
3. Use generic messages for unexpected errors

Security Considerations

1. Don't Expose Sensitive Information

```
// Good - Generic message throw new VehicleNotFoundException(
messageSourceHelper.getMessage("vehicle.notFound", id)); // Avoid - Reveals database
structure throw new Exception("SELECT * FROM vehicles WHERE id = " + id + " returned no
rows");
```

2. Log Security Events

```
@ExceptionHandler(InvalidTokenException.class) public ErrorResponse
handleInvalidToken(InvalidTokenException ex) { log.warn("Invalid token attempt from IP: {}",
getClientIP()); // Return generic message }
```

3. Rate Limiting for Auth Errors

Implement rate limiting for repeated authentication failures.

Migration Summary

Custom Exceptions Created: 50+

GlobalExceptionHandler: Complete coverage

HTTP Status Codes: Properly mapped

i18n Integration: 100% translated

Validation: Bean Validation + Custom

Result: Consistent error handling across entire API 

Conclusion

The exception handling implementation provides a robust, consistent, and user-friendly error response system. Integration with i18n ensures localized error messages, while proper HTTP status codes enable correct client-side handling.

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